

Standard Operating Procedure ***Metal Shear***

Metal Shear

A metal shear is a high-force machine designed to cut sheet metal and plate stock to size using a heavy-duty blade and a fixed bed. It is an essential tool for rapid stock preparation but presents significant mechanical hazards due to the immense force required for shearing operations..

Potential Hazards

- Be aware of the following potential hazards when working with Horizontal Band saws:
  - **Mechanical:** points of operation, pinch points, shear points, power transmission points
  - **Operational:** heat, dust/chips/shards, noise, unstable loads/stocks
  - **Electrical/Energy Sources:** exposed wiring, malfunctioning equipment, unexpected start up/shut down, hydraulic power.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**Safety Glasses (ANSI Z87.1):** Required at all times to protect against metal shards and projectile debris.
Footwear	**Closed-Toe Shoes:** Steel-toed boots are recommended when handling heavy plate stock or large sheet material.
Hand Protection	**Heavy-Duty Work Gloves:** Advised when handling raw stock to prevent lacerations from sharp "burred" edges.
Clothing/Hair	Tie back long hair and secure loose clothing. Remove all jewelry, especially rings, which can be caught in material during the feed.



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Before Use

- **Workspace Inspection**: Ensure the area around the shear is clear of off-cuts, tools, and tripping hazards.
- **Machine Guarding**: Verify that the finger guard is in place and adjusted to prevent hands from entering the blade's travel path.
- **Capacity Verification**: Confirm the material thickness and type do not exceed the machine's rated capacity (e.g., 1/4in mild steel, 1/8th in stainless). Never attempt to shear hardened steel or round stock.
- **Blade Check**: Inspect the blade for nicks or dullness. Report any damage to the shop supervisor immediately.
- **Back Gauge Setup**: Adjust the back gauge to the required dimension before loading material.

During Use

- **Concentration**: Avoid distractions; injuries on shears often occur due to a lack of focus on hand placement.
- **Material Alignment**: Hold the stock flat against the bed and firmly against the side square.
- **The "4-Inch" Rule**: Keep hands at least 4 inches away from the blade and hold-down area at all times. Use a push board for small pieces.
- **Hold-Downs**: Ensure the automatic or manual hold-downs are engaged before the blade begins its descent to prevent material "kick-up."
- **One Operator**: Only one person should control the machine. If a second person is needed to help support large sheets, they must be clear of all moving parts before the cut is initiated.



- **Treadle Safety**: Only place your foot on the treadle when you are ready to make the cut. Remove your foot immediately after the cycle is complete.

After Use

- Stock will be sharp around the cut face.
- Sweep or use a shop vacuum to collect any debris that resulted from cutting on the floor or other surfaces. Avoid cleaning with compressed air to prevent injury.
- Store stock materials in a neat and secured manner; do not accumulate excess combustibles. Keep aisles and exits clear.

Emergency Equipment and Procedures

- Report all incidents to the shop supervisor, including injuries, equipment damage, and near misses.
- Know the location of the nearest emergency equipment and items, such as the emergency stop/shut-off for the specific machine, eye wash, first aid kit, fire extinguishers, fire alarm pull stations, and emergency exits.